REPORT

Telecom Customer Churn Analysis & Prediction

From Data to Decision: Solving a Real-World Business Problem with Data Science

Dataset	IBM Telco Customer Churn: Kaggle (7,043 customers, 21 features)
Tools Used	Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn) · Tableau · Google Colab
Date	June 2026

1. Project Overview

This capstone project analyses customer churn behaviour for a telecommunications company using real-world data. The project follows a complete, end-to-end data science workflow, from raw data ingestion through cleaning, exploratory analysis, machine learning modelling, and stakeholder-ready visualisation.

1.1 Business Problem

Customer churn, the loss of subscribers to competitors or cancellation is one of the most critical challenges in the telecommunications industry. Acquiring a new customer costs significantly more than retaining an existing one. Every churned customer represents lost recurring revenue and increased customer acquisition cost.

This project addresses the following core business question:

> ***"What factors drive customer churn, and can we predict which customers are likely to leave next?"***

1.2 Project Objectives

- Understand the profile of customers who churn versus those who stay
- Identify the strongest predictors of churn from available features
- Build a machine learning model that can flag at-risk customers
- Deliver actionable recommendations to reduce churn
- Communicate findings through an interactive Tableau dashboard

7,043	26.5%	1,869	21
Total Customers	Churn Rate	Churned Customers	Features Analysed

2. Dataset Description

The dataset used is the IBM Telco Customer Churn dataset, publicly available on Kaggle. It contains billing and service information for 7,043 telecom customers, each labelled as churned (Yes) or retained (No).

2.1 Dataset Source

- **Kaggle: IBM Telco Customer Churn** Source:

- https://www.kaggle.com/datasets/blastchar/telco-customer-churn
- Rows: 7,043 customer records
- Columns: 21 features (20 predictors + 1 target variable)

2.2 Key Features

Column	Type	Description
tenure	Numeric	Number of months the customer has been with the company
Contract	Categorical	Month-to-month, One year, or Two year contract
MonthlyCharges	Numeric	The amount charged to the customer each month
TotalCharges	Numeric	Total amount charged (stored as string, required cleaning)
InternetService	Categorical	DSL, Fiber optic, or No internet service
PaymentMethod	Categorical	Electronic check, Mailed check, Bank transfer, Credit card
SeniorCitizen	Binary	1 if senior citizen, 0 if not (stored as integer, required conversion)
Churn	Target	Yes = customer left, No = customer stayed

3. Data Wrangling & Cleaning

The raw dataset required several cleaning steps before analysis could begin. This section documents every transformation applied and the reasoning behind each decision.

Issue Found	Action Taken	Reason
TotalCharges stored as string with 11 blank values	Converted to float using pd.to_numeric(errors='coerce'), filled blanks with 0	Blank values belonged to new customers (tenure=0) with no charges yet
SeniorCitizen stored as 0/1 integer	Mapped to 'Yes'/'No' string labels	Consistent with all other binary columns for readability and Tableau compatibility
customerID column present	Dropped from dataset	Unique identifier, adds no predictive or analytical value
No duplicate rows detected	No action required	Verified with df.duplicated().sum() returning 0
Churn column as Yes/No string	Encoded to binary (1/0) for modelling	Required for correlation analysis and classification models

No tenure groups present	Engineered tenure_group feature (0-12, 13-24, 25-48, 49-72 months)	Enables richer EDA and more meaningful Tableau visualisations

4. Exploratory Data Analysis

Ten visualisations were produced to explore patterns in the data. Each chart was designed to answer a specific business question.

Chart	Key Finding	Business Implication
Overall Churn Distribution	26.5% of customers churned (1,869 of 7,043)	1 in 4 customers is leaving, significant revenue risk
Churn by Contract Type	Month-to-month: 42.7% churn vs 2.8% for two-year	Contract length is the single strongest retention lever
Churn by Tenure Group	0-12 months: 47.4% churn; 49-72 months: 9.5%	Early loyalty is critical, focus on first-year retention
Churn by Internet Service	Fiber optic: 41.9% vs DSL: 19% vs No internet: 7.4%	Premium service has highest dissatisfaction, pricing or quality issue
Monthly Charges vs Churn	Churned customers paid avg \$74.44 vs \$61.27 retained	Price sensitivity is a churn driver
Churn by Payment Method	Electronic check: 45.3% vs Credit card auto: 15.2%	Auto-pay users are far more loyal
Churn by Gender	Male and female churn rates nearly identical	Gender is not a useful segmentation variable for retention
Senior Citizen Churn	Senior citizens churn at ~41% vs ~24% non-seniors	Senior segment needs dedicated retention support
Value-Added Services vs Churn	Customers without Online Security or Tech Support churn more	These services create stickiness, promote them actively
Correlation Heatmap	Tenure, Contract, MonthlyCharges most correlated with Churn	Confirms feature selection for modelling

5. Machine Learning Modelling

5.1 Preprocessing Steps

- Label encoded all categorical variables using sklearn LabelEncoder
- Addressed class imbalance using SMOTE, upsampled minority class (churned) to match majority
- Applied 80/20 train-test split with stratification to preserve class ratios
- Scaled all features using StandardScaler (mean=0, std=1) for Logistic Regression

5.2 Models Trained

- Logistic Regression: interpretable baseline model
- Random Forest Classifier: ensemble model, 100 trees

5.3 Model Performance

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	~80%	~79%	~80%	**~0.88**
Random Forest (Selected)	~85%	~85%	~85%	**~0.93**

The Random Forest model was selected as the final model due to its superior ROC-AUC score of ~0.93, indicating strong ability to distinguish between churned and retained customers. Five-fold cross-validation was applied to verify result stability.

5.4 Most Important Features (Random Forest)

- tenure: length of customer relationship is the strongest predictor
- MonthlyCharges: higher charges correlate strongly with churn risk
- TotalCharges: cumulative spend reflects long-term relationship value
- Contract: contract type is the most actionable business lever
- InternetService: service type significantly influences churn probability

6. Key Insights

Critical Risk Factors

- Month-to-month customers churn at 42.7%: nearly 15x the rate of two-year contract customers (2.8%)
- New customers (0-12 months) are the highest-risk group at 47.4% churn rate
- Fiber optic users churn at 41.9% despite paying the most: suggesting value dissatisfaction
- Electronic check users churn at 45.3%: the highest of all payment methods
- Senior citizens churn at approximately 41% compared to 24% for non-seniors

Moderate Risk Factors

- Customers paying above-average monthly charges ($74.44 avg for churned vs $61.27 for retained) show higher price sensitivity
- Customers without Online Security or Tech Support services are significantly more likely to churn
- Customers without partners or dependents churn more: family accounts are stickier

Retention Strengths

- Long-tenured customers (49+ months) are extremely loyal at only 9.5% churn rate
- Auto-payment users (bank transfer and credit card) are the most retained segments
- Two-year contract customers almost never churn (2.8%): contracts create commitment

7. Business Recommendations

#	Recommendation	Action	Expected Impact
1	**Incentivise longer contracts**	Offer 10-20% discounts to customers who upgrade from monthly to annual contracts, especially in first 6 months	Could reduce monthly churn rate by up to 15%
2	**Improve early onboarding (0-12 months)**	Launch a 90-day onboarding program with proactive check-ins, personalised offers, and dedicated new-customer support	Target the highest-risk cohort directly at their peak churn window
3	**Investigate Fiber Optic pricing and quality**	Conduct customer satisfaction surveys among Fiber Optic users. Consider a price-lock guarantee for year one.	Fiber optic customers are highest value: retaining them has maximum revenue impact
4	**Promote auto-pay enrolment**	Offer $5/month discount for switching to automatic bank transfer or credit card payment	Auto-pay users churn at only 15-17% vs 45% for electronic check users
5	**Deploy churn prediction model in CRM**	Flag customers with >70% predicted churn probability and trigger a retention offer 30 days before contract end	Proactive intervention before churn occurs — shift from reactive to predictive retention

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8. Tableau Dashboard

An interactive Tableau dashboard was built using the cleaned dataset exported from the Python notebook. The dashboard tells the complete churn story visually and allows stakeholders to explore the data interactively.

8.1 Dashboard Components

- KPI Overview: headline churn rate (26.5%) and total churned customers (1,869)
- Churn by Contract Type: horizontal bar chart showing month-to-month dominance
- Churn by Tenure Group: colour-coded bar chart from red (high risk) to green (loyal)
- Churn Rate by Internet Service: with average reference line
- Monthly Charges vs Churn: side-by-side bars showing price differential
- Churn by Payment Method: Electronic check flagged at 45.3%

8.2 Interactive Filters

- Gender, Senior Citizen, Contract Type, Internet Service, Partner. all cross-filter the entire dashboard simultaneously

8.3 File

- Tableau workbook submitted as: telecom_churn_dashboard.twbx (data embedded)

9. Conclusion

This capstone project successfully demonstrated a complete, reproducible data science workflow applied to a real-world business problem. Starting from a raw, uncleaned dataset, the project progressed through systematic data wrangling, thorough exploratory analysis, machine learning modelling with proper validation, and stakeholder-ready visual communication.

The Random Forest model achieved approximately 85% accuracy and a ROC-AUC of ~0.93, providing strong predictive power to identify at-risk customers before they churn. Combined with the five evidence-based business recommendations, this project delivers both analytical rigour and practical business value.

> **Final Answer:** *Contract type, customer tenure, internet service type, and payment method are the strongest predictors of churn. Our model can identify at-risk customers with ~93% AUC, enabling the business to intervene proactively shifting retention strategy from reactive to predictive.*